

```
import pandas as pd

# Paste your raw GitHub URL here
url = 'https://raw.githubusercontent.com/SMont84/ITAI-1371-Midterm-GitGat/refs/heads/main/data/stroke_data.csv'

df = pd.read_csv(url)
df.head()
```

	id	gender	age	hypertension	heart_disease	ever_married	work_type
0	9046	Male	67.0	0	1	Yes	Private
1	51676	Female	61.0	0	0	Yes	Self-employed
2	31112	Male	80.0	0	1	Yes	Private
3	60182	Female	49.0	0	0	Yes	Private
4	1665	Female	70.0	1	0	Yes	Self-employed

```
!pip install -q imbalanced-learn
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from imblearn.over_sampling import SMOTE
from sklearn.preprocessing import LabelEncoder

target_column = 'stroke'

# 1. Convert target text to numbers if it's text
le = LabelEncoder()
df[target_column] = le.fit_transform(df[target_column])

# 2. Separate X and y
X = df.drop(columns=[target_column])
y = df[target_column]

# 3. Convert feature text columns to numbers
X = pd.get_dummies(X)

# 4. Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
print(df.isna().sum())
```

```
id          0
gender      0
```

```

gender          0
age             0
hypertension    0
heart_disease   0
ever_married    0
work_type       0
Residence_type  0
avg_glucose_level 0
bmi            201
smoking_status  0
stroke          0
dtype: int64

```

```
df = df.fillna(df.mean(numeric_only=True))
```

```
print(X.isna().sum())
```

```

id                0
age              0
hypertension      0
heart_disease     0
avg_glucose_level 0
bmi             201
gender_Female     0
gender_Male       0
gender_Other      0
ever_married_No   0
ever_married_Yes  0
work_type_Govt_job 0
work_type_Never_worked 0
work_type_Private 0
work_type_Self-employed 0
work_type_children 0
Residence_type_Rural 0
Residence_type_Urban 0
smoking_status_Unknown 0
smoking_status_formerly smoked 0
smoking_status_never smoked 0
smoking_status_smokes 0
dtype: int64

```

```
import numpy as np
```

```

# Count how many infinite values exist in each column
print(np.isinf(X).sum())

```

```

id                0
age              0
hypertension      0
heart_disease     0
avg_glucose_level 0
bmi             0
gender_Female     0

```

```

gender_Male          0
gender_Other         0
ever_married_No      0
ever_married_Yes     0
work_type_Govt_job   0
work_type_Never_worked 0
work_type_Private    0
work_type_Self-employed 0
work_type_children   0
Residence_type_Rural 0
Residence_type_Urban 0
smoking_status_Unknown 0
smoking_status_formerly smoked 0
smoking_status_never smoked 0
smoking_status_smokes 0
dtype: int64

```

```

from sklearn.impute import SimpleImputer

imputer = SimpleImputer(strategy='mean')
X_train_imputed = imputer.fit_transform(X_train)
X_test_imputed = imputer.transform(X_test)

smote = SMOTE(random_state=42)
X_train_resampled, y_train_resampled = smote.fit_resample(X_train_imputed, y

# 10. Verify the results
print("Original Training Shape:", X_train_imputed.shape)
print("Resampled Training Shape (After SMOTE):", X_train_resampled.shape)
print("Success! SMOTE completed.")

```

```

Original Training Shape: (4088, 22)
Resampled Training Shape (After SMOTE): (7778, 22)
Success! SMOTE completed.

```

```

# Identify any columns containing text
text_columns = df.select_dtypes(include=['object', 'string']).columns
print("Columns with text:", text_columns)

```

```

Columns with text: Index(['gender', 'ever_married', 'work_type', 'Residence_t
'smoking_status'],
dtype='object')

```

```

import pandas as pd

# 1. Create the encoded dataframe
df_encoded = pd.get_dummies(df)

# 2. Print the new columns to confirm 0s and 1s appear
print(df_encoded.head())

```

```
# 3. Export the encoded version
```

```
df_encoded.to_excel('fully_encoded_dataset.xlsx', index=False)
```

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	\
0	9046	67.0	0	1	228.69	36.600000	
1	51676	61.0	0	0	202.21	28.893237	
2	31112	80.0	0	1	105.92	32.500000	
3	60182	49.0	0	0	171.23	34.400000	
4	1665	79.0	1	0	174.12	24.000000	

	stroke	gender_Female	gender_Male	gender_Other	...	\
0	1	False	True	False	...	
1	1	True	False	False	...	
2	1	False	True	False	...	
3	1	True	False	False	...	
4	1	True	False	False	...	

	work_type_Never_worked	work_type_Private	work_type_Self-employed	\
0	False	True	False	
1	False	False	True	
2	False	True	False	
3	False	True	False	
4	False	False	True	

	work_type_children	Residence_type_Rural	Residence_type_Urban	\
0	False	False	True	
1	False	True	False	
2	False	True	False	
3	False	False	True	
4	False	True	False	

	smoking_status_Unknown	smoking_status_formerly smoked	\
0	False	True	
1	False	False	
2	False	False	
3	False	False	
4	False	False	

	smoking_status_never smoked	smoking_status_smokes
0	False	False
1	True	False
2	True	False
3	False	True
4	True	False

```
[5 rows x 23 columns]
```

```
# This lists all variables and checks if they are pandas DataFrames
for name, val in list(globals().items()):
```

```
    if isinstance(val, pd.DataFrame):
```

```
        print(f"DataFrame Name: '{name}' | Shape: {val.shape}")
```

```
DataFrame Name: '_' | Shape: (5, 12)
```

```
DataFrame Name: 'df' | Shape: (5110, 12)
```

```
DataFrame Name: '_1' | Shape: (5, 12)
DataFrame Name: 'X' | Shape: (5110, 22)
DataFrame Name: 'X_train' | Shape: (4088, 22)
DataFrame Name: 'X_test' | Shape: (1022, 22)
DataFrame Name: 'df_encoded' | Shape: (5110, 23)
```

```
# 1. Convert True/False values to 1 and 0
df_encoded_numeric = df_encoded.replace({True: 1, False: 0})

# 2. Save the converted DataFrame to an Excel file
df_encoded_numeric.to_excel("scrubbed_dataset_encoded.xlsx", index=False)

# 3. Download the file to your computer
from google.colab import files
files.download("scrubbed_dataset_encoded.xlsx")
```

```
/tmp/ipykernel_1129/1888596896.py:2: FutureWarning: Downcasting behavior in `
df_encoded_numeric = df_encoded.replace({True: 1, False: 0})
```